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END SEMESTER EXAMINATION 2025

Name of the Course: B.TECH
CSE

Semester: VI _____

Name of the Paper: Full Stack
Web Development

Paper Code: TCS 693

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Explain some differences between HTML-4 and HTML-5. Can you state a few differences along with the explanation of audio and video tags of HTML-5 with its various attributes? Write down a JavaScript code that controls the play and pause of audio and video on a page using a toggle button.	CO1,CO2, CO3
(b)	What are the different methods available for including CSS in an HTML document? Please describe all the possible ways to incorporate CSS into HTML and discuss the advantages and disadvantages of each method.	
(c)	Explain the difference between GET and POST method. Discuss the use of event listeners in JavaScript to capture form submission and manipulate form data (e.g., preventing default form submission, dynamically changing form content).	
Q2	(20marks)	
(a)	Define XML and list the rules to be followed by XML documents What is a PCDATA and CDATA section in DTD? With the help of XML, create a database of Courses and validate it with the help of DTD	CO3,CO4
(b)	Explain the concept of AJAX? Write a program to fetch the content of a JSON file named json_demo.json with the help of AJAX and visualize the database on the client side. <pre> { "name": "John", "birth": "1986-09-11", "age": "function(){ return 34;}", "pets": [{ "animal": "dog", "name": "Fido" }, { "animal": "cat", "name": "Felix" }, { "animal": "hamster", "name": "Lightning" }] } </pre>	
(c)	Explain the difference between JavaScript and jQuery? What are the two ways of using jQuery on your website, and which one of them is better, and why? Please write the syntax of jQuery and explain how you can hide and unhide an element (using element tag, class, and ID for selecting element) when the mouse enters or leaves that element.	
Q3	(20marks)	
		CO4

(a)	What is an associative array in PHP and how does it differ from an indexed array? Explain a few functions for sorting in associative arrays in PHP and also write PHP script to create an associative array of student names and their marks. Sort the array by values in ascending order and by keys in descending order.	
(b)	Describe how to establish a connection to a MySQL database using PHP with mysqli or PDO. Write PHP scripts to insert , update , and delete records in a students table with the following fields: EnrolmentNo, Username, Name, Email, PhoneNo, Gender, Country.	
(c)	What is the difference between sessions and cookies in PHP? Compare them in terms of storage location, security, and typical use-cases. How data can be stored and retrieved using sessions and cookies in PHP and also explain how session data is destroyed and how cookies can be deleted from the client browser.	
Q4	(20 marks)	
(a)	Describe the typical steps involved in reading from and writing to a file using PHP. Write a PHP script that creates a text file named log.txt, writes a message to it, and then reads and displays its contents on the browser.	CO3, CO4, CO5
(b)	Describe the steps involved in securely uploading a file to the server and checking for potential security risks like file size, type, and extension. What is the role of the \$_FILES superglobal?	
(c)	What is Cross-Site Scripting (XSS)? Explain the different types of XSS attacks (Stored, Reflected, and DOM-based) with examples. Write a PHP program that includes a form and safely displays submitted user input. Use htmlspecialchars() and \$_SERVER['PHP_SELF'] to demonstrate XSS prevention on the client/server side.	
Q5	(20 marks)	
(a)	Analyze the role of CMS in modern web development. Compare WordPress with at least two other popular CMS platforms in terms of features, flexibility, and use-cases. From a technical perspective, explain how WordPress handle's themes, plugins, and database interactions	CO4, CO5
(b)	Differentiate between client-side and server-side validation with examples. Which one is more secure, and why? Write a code snippet that validates an email address and mobile number using regular expressions on server side.	
(c)	Explain the components of the MERN stack (MongoDB, Express.js, React.js, and Node.js) and how they interact in a typical full-stack web application. Illustrate the request-response lifecycle through the stack with a simple use-case like user login or data submission.	

Note for the question paper setters:

- Question paper should cover all the COs of the course.